

# Implementing OpenLDAP Directory Services



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# Overview



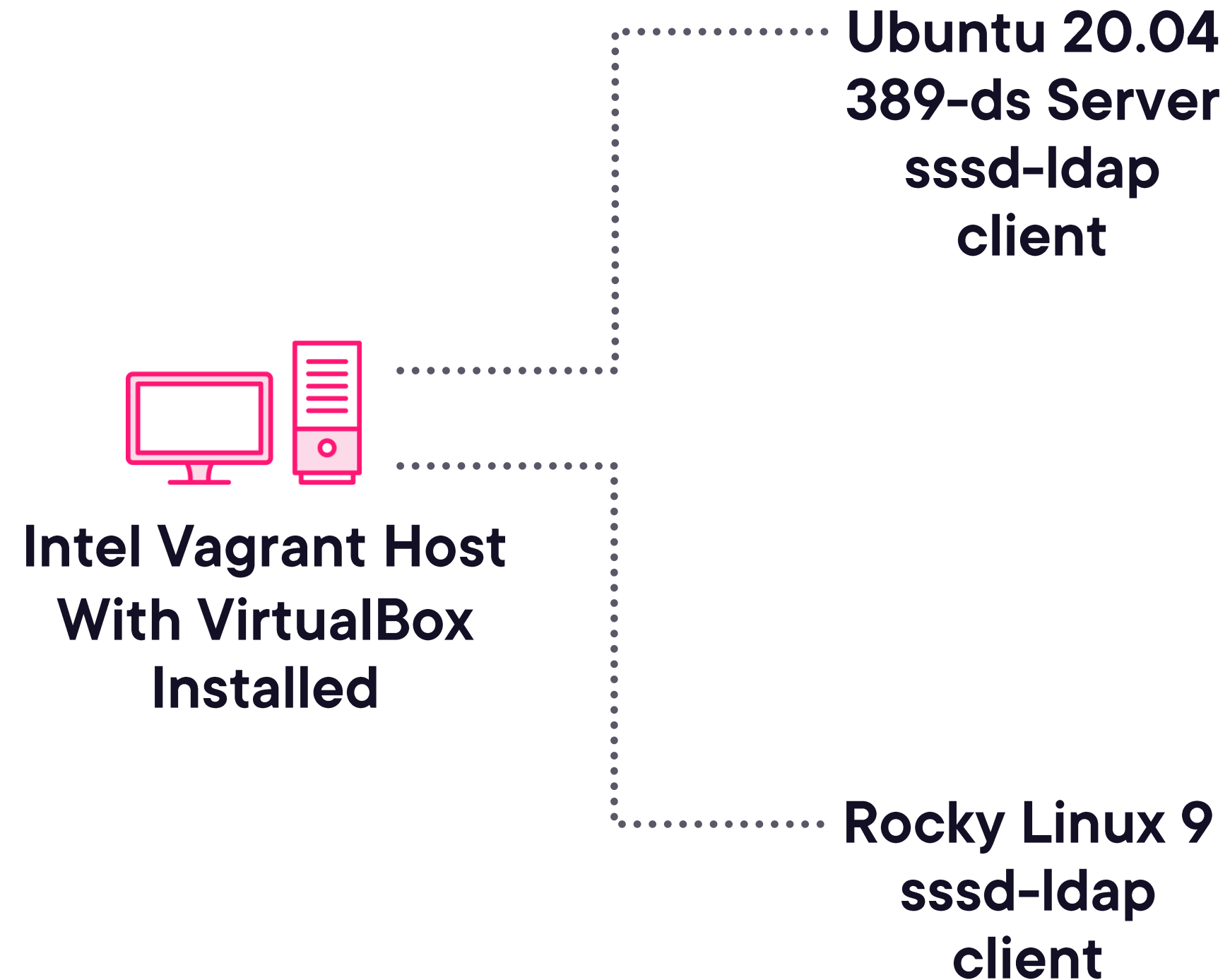
## Directory Services

- Install and configure 389-ds on Ubuntu
- Using LDAP tools to query directory
- Create LDAP accounts
- Configure sssd-ldap client on Ubuntu
- Configure sssd-ldap client on Rocky Linux



# Lab Systems

<https://github.com/theurbanpenguin/lfcs>





# Centralized User Accounts



# Local vs Centralized

**Local**

**/etc/passwd**

**Directory**

**OpenLDAP / Active Directory**



# 389-ds



## Core version of Red Hat Directory Services

- OpenLDAP based
- Simplified management and setup
- Configured for clear text 389 LDAP and TLS 636 LDAPS
- LDAPS uses self-signed certificate by default
- Installing and configuring OpenLDAP is not an exam objective, but you need to connect to OpenLDAP



# Demo



## Install and configure 389-ds:

- Installing 389-ds does not configure the directory
- One or more directory services can be configured with dscreate





# LDAP Tools



```
# ldapsearch -H ldap://ubuntu1:389 -D 'cn=Directory Manager' -W -b 'dc=example,dc=com' -x
# ldapsearch -H ldaps://ubuntu1:636 -D 'cn=Directory Manager' -W -b 'dc=example,dc=com' -x
# echo 'TLS_REQCERT never' >> /etc/ldap/ldap.conf
# ldapsearch -H ldaps://ubuntu1:636 -D 'cn=Directory Manager' -W -b 'dc=example,dc=com' -x
```

## **389 or 636 / LDAP or LDAPS**

**LDAPS is preferred but the standard tools will complain about self-signed certificates. If you know and trust the self-signed certificate, you can disable the check**



```
# vim ~/.dsrc
[localhost]
uri = ldapi://%%2fvar%%2frun%%2fslapd-localhost.socket
basedn = dc=example,dc=com
binddn = cn=Directory Manager
# dsidm localhost user list
```

## 389-ds Tools

LDAP management is simplified using 389-ds. Configuring the .dsrc allows for common settings to be stored. When using a forward-slash in the URI, make sure %%2f is used:

%% = percent encoding  
2F = ASCII 47 (/) in HEX



```
# dsidm localhost user create --uid Alice \  
  --cn "Alice User" --displayName "Alice User" \  
  --uidNumber 2000 --gidNumber 100 \  
  --homeDirectory /home/alice  
# dsidm localhost account reset_password "uid=alice,ou=people,dc=example,dc=com"
```

## Create User

**Creating a user account is easy. When setting the password, the account FQDN must be used.**



# Demo



**Using Directory Tools:**

- Cockpit Web Manager



# Demo



## Using Directory Tools:

- ldapsearch
- /etc/ldap/ldap.conf
- dsidm





# **Configure Authentication: Ubuntu**



# Directory Service

## White Pages

Used to search for employees  
or entries of some sort

## Authentication

Can be used to authenticate  
Linux users or others



# User Home Directories

## Centralized NFS Share

Ideally, centralized user's have centralized home directories. This enables standardized content each time they login from any server. Mount the `/export/home` directory or which ever mountpoint you want to the NFS exported directory

## Create Home Directory on Login

Using `oddjob` or `pam_mkhomedir` a home directory can be created on user logon if it does not exist. Creating a user in the Directory is NOT going to create the home directory for the user even on the Directory server



```
# apt install -y sssd-ldap ldap-utils
# vim /etc/sss/sss.conf
# chmod 600 /etc/sss/sss.conf
# systemctl restart sssd
# systemctl status sssd
# getent passwd alice
# echo "session optional pam_mkhomedir.so" >> /etc/pam.d/common-session
$ su - alice
```

## Install and Configure SSSD LDAP Authentication

The System Security Services Daemon is preferred to authentication. It can allow for more than one LDAP directory to be queried. The sssd.conf file is available in the course git repo



# Demo



## Configure Authentication:

- Working on Ubuntu we configure LDAP authentication



# **Configure Authentication: Rocky 9**





# Shared Accounts

The big advantage of using a Directory Service is that account details can be shared between systems



```
$ sudo dnf -y install openldap-clients sssd sssd-ldap oddjob-mkhomedir vim
$ authselect list
$ sudo authselect select sssd with-mkhomedir --force
$ sudo systemctl enable --now oddjobd.service
$ sudo vim /etc/sss/sss.conf
# sudo chmod 600 /etc/sss/sss.conf
# sudo systemctl restart sssd
```

## Working with Rocky 9

The process is similar on Rocky 9. The same sss.conf can be used. The oddjob daemon is preferred for creating home directories



# Demo



## Configure Authentication:

- Working on Rocky we configure LDAP authentication



**Stop Right There!**



```
$ sudo -i  
# dsidm localhost user get alice  
# dsidm localhost user modify alice add:loginShell:/bin/bash
```

## Consistency

**The default shell may not be consistent between your Linux guests. To ensure consistency you can configure the shell in the Directory**



# Demo



## Modify Users:

- Configure Login Shell



# Summary



## Directory Services

- OpenLDAP
- Active Directory

## 389-ds

- Wrapper to OpenLDAP
- 389 LDAP
- 636 LDAPS

## Home Directories

- Mount /export/home
- Oddjob
- pam\_mkhomedir

**Up Next:**

**Congratulations**

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